

09/09/2025: Boolean Algebra

CSCI 246: Discrete Structures

Textbook reference: Sec. 7, Scheinerman

Quiz return method

Quizzes are grouped into four bins (A-G, H-L, M-R, S-Z) by last name. The quizzes are upside down with your last name on the back. Come find yours before, during, or after class. Only turn the quiz over if it's yours.

Today's Agenda

- Quiz updates (≈ 5 mins)
- Review counterexamples (≈ 15 mins)
- Boolean algebra (≈ 15 mins)
- Group exercises (≈ 15 mins)

Quiz Updates

Review of Counterexamples

Boolean Algebra

Weekly Quiz #1: Scores

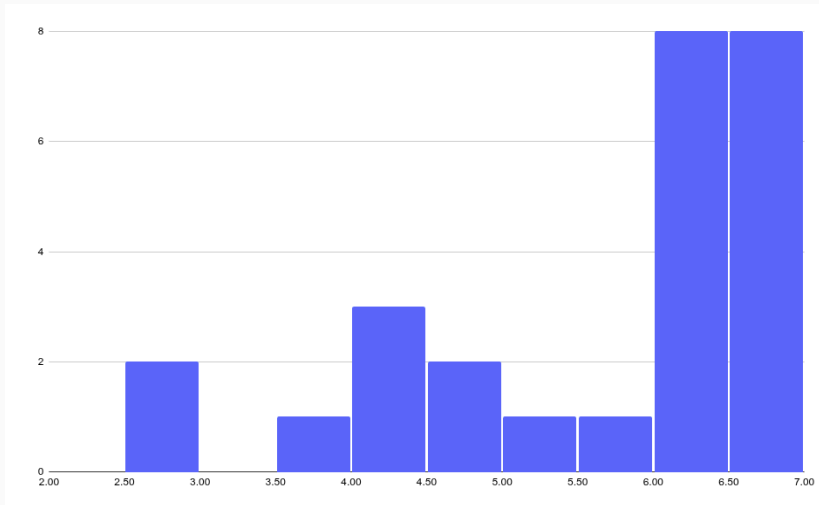


Figure 1: Reading Quiz Scores. Median: 6/7 (85.7%). See repo for feedback.

Study Guide For Quiz 2

Readings (1 question)

Sec 5 (Proofs)

- Know how to prove Props 5.2, 5.3, 5.5, and 5.6.

Sec. 6 (Counterexample)

- Know how to apply Proof Template 3.
- In particular, know how to disprove Statement 6.1.

Group exercises (2 questions)

Know how to do all of the group exercises from these 2 sections.

Quiz Updates

Review of Counterexamples

Boolean Algebra

Review of Sec. 6 (Counterexamples)

Disprove the following conjecture: *Let a and b be integers. If $a|b$ and $b|a$, then $a = b$.*

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Is $a = 1, b = 0$ a valid counterexample?

Review of Sec. 6 (Counterexamples)

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Poll

Is $a = 1, b = 0$ a valid counterexample?

Recall. Here's the truth table for unidirectional implication.

		If X, then Y $X \Rightarrow Y$	
X	Y		
T	T	T	
T	F	F	← counterexamples live here
F	T	T	
F	F	T	

Review of Sec. 6 (Counterexamples)

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Recall. Here's the truth table for unidirectional implication.

		If X , then Y $X \implies Y$	
X	Y	$X \implies Y$	
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T	F	F	← counterexamples live here
F	T	T	
F	F	T	

Interpretation. So we need to find a case where proposition $X = \{a|b \text{ and } b|a\}$ is true but proposition $Y = \{a = b\}$ is false.

Review of Sec. 6 (Counterexamples)

Disprove the following conjecture: *Let a and b be integers. If $a|b$ and $b|a$, then $a = b$.*

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Reminder of Definition 3.2 (**Divisible**)

Let a and b be integers. We say that a is *divisible* by b provided there is an integer c such that $bc = a$. The notation for this is $b|a$.

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Solution to poll: The proposition $a|b$ ($= 1|0$) is true, since there is an integer c (namely $c = 0$) such that $ac = b$ (since $1 \cdot 0 = 0$). However, the proposition $b|a$ ($= 0|1$) is false, since there is no integer d such that $bd = a$ (that is, there is no integer d such that $0 \cdot d = 1$). Overall, **the hypothesis doesn't hold, so we have not disproven the statement.**

Review of Sec. 6 (Counterexamples)

Disprove the following conjecture: *Let a and b be integers. If $a|b$ and $b|a$, then $a = b$.*

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Main Text

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Annotation

High-level structure

Main Text

Let $a = 5, b = -5$. First, we show that the hypothesis holds [i.e., that $(5|-5)$ and $(-5|5)$].

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Unravel defn.

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Let $a = 5, b = -5$. First, we show that the hypothesis holds [i.e., that $(5|-5)$ and $(-5|5)$].

$a|b$ means there is an integer x such that $ax = b$. Likewise $b|a$ means there is an integer y such that $by = a$.

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Let $a = 5, b = -5$. First, we show that the hypothesis holds [i.e., that $(5|-5)$ and $(-5|5)$].

$a|b$ means there is an integer x such that $ax = b$. Likewise $b|a$ means there is an integer y such that $by = a$.

Substituting for a and b , we need to show that there are integers x and y such that $5x = -5$ and $-5y = 5$. We see these equations hold by taking $x = -1$ and $y = -1$.

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Hence $5|-5$ and $-5|5$, so the hypothesis is met.

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Now we show that the conclusion fails [i.e. that $-5 \neq 5$.]

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Hence $5|-5$ and $-5|5$, so the hypothesis is met.

Now we show that the conclusion fails [i.e. that $-5 \neq 5$.]

This is immediately clear.

Review: how to give a counterexample

Unidirectional implication: $X \implies Y$

Exhibit **one** case in which

- the hypothesis X **holds**, and
- the conclusion Y **fails**.

To check either one, you must **unravel the definitions** inside X and Y .

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What do you have to show to give a counterexample to a bidirectional implication: a proposition of the form $X \iff Y$?

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Poll

What do you have to show to give a counterexample to a bidirectional implication: a proposition of the form $X \iff Y$?

Solution

Show that **either** direction fails: **either** that $X \implies Y$ fails, **or** that $Y \implies X$ fails.

Poll

Is the following statement true or false?

$$-5|5 = -1$$

A note

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Solution to poll: The statement is **incorrect**! Why?

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Solution to poll: The statement is **incorrect**! Why? $-5|5$ is a **proposition**, so it can't equal -1 . We would instead write just $-5|5$, or perhaps $-5|5 = \text{TRUE}$.

Quiz Updates

Review of Counterexamples

Boolean Algebra

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What is algebra?

Algebra lets us reason about **numbers**. For example, using algebra, we can show that

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What is Boolean algebra?

Boolean Algebra lets us reason about **propositions**. For instance, using Boolean Algebra, we can show that

- If A, then B
- (not A) or B

mean the same thing.

Reasoning about propositions

To reason about propositions, we can use two tools:

- Truth Tables
- Boolean Algebra Properties

Tool #1: Truth Tables

Practice Reading Quiz (Sec. 7 - Boolean Algebra)

Use a truth table to prove that the expressions $x \implies y$ and $(\neg x) \vee y$ are logically equivalent.

Notation reminder

- $\implies$ means implies
- $\neg$ means not
- $\vee$ means or

Tool #1: Truth Tables

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Solution. The 3rd and 6th columns are the same.

X	Y	$X \implies Y$	$\neg X$	Y	$(\neg X) \vee Y$
T	T	T	F	T	T
T	F	F	F	F	F
F	T	T	T	T	T
F	F	T	T	F	T

Tool #2: Properties

In the group exercises, we will see that we can also use the following properties to reason about propositions.

Theorem 7.2

- $x \wedge y = y \wedge x$ and $x \vee y = y \vee x$. (Commutative properties)
- $(x \wedge y) \wedge z = x \wedge (y \wedge z)$ and $(x \vee y) \vee z = x \vee (y \vee z)$. (Associative properties)
- $x \wedge \text{TRUE} = x$ and $x \vee \text{FALSE} = x$. (Identity elements)
- $\neg(\neg x) = x$.
- $x \wedge x = x$ and $x \vee x = x$.
- $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$ and $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$. (Distributive properties)
- $x \wedge (\neg x) = \text{FALSE}$ and $x \vee (\neg x) = \text{TRUE}$.
- $\neg(x \wedge y) = (\neg x) \vee (\neg y)$ and $\neg(x \vee y) = (\neg x) \wedge (\neg y)$. (DeMorgan's Laws)

Figure 2: Boolean Algebra Properties

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These properties are proven via truth tables. You can use them as **shortcuts**.



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Boolean Algebra Properties: Example

DeMorgan's Laws

For propositions P and Q :

$$1. \quad \neg(P \wedge Q) = (\neg P) \vee (\neg Q)$$

$$2. \quad \neg(P \vee Q) = (\neg P) \wedge (\neg Q)$$

Poll

How can we understand/interpret these?

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Poll

How can we understand/interpret these?

Verbalization

1. "Not (P and Q)" is the same as "(Not P) or (Not Q)."
2. "Not (P or Q)" is the same as "(Not P) and (Not Q)."

Boolean Algebra Properties: Example

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For propositions P and Q :

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People use these all of the time in conversation!

Law 1 “You can’t **have your cake** **and** **eat it too**.”

Means: either you don’t **have it**, **or** you don’t **eat it**. That is $\neg(P \wedge Q) = (\neg P) \vee (\neg Q)$.

Law 2 “I don’t **drink** **or** **smoke**.”

Nobody hears that as “or”. Everyone hears “I don’t **drink** **and** I don’t **smoke**”. That is, $\neg(P \vee Q) = (\neg P) \wedge (\neg Q)$, applied without noticing.

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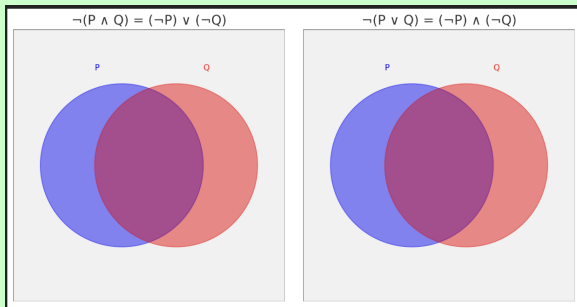
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Visual intuition



- On the left, $\neg(P \wedge Q)$: everything except the overlapping middle part.
- On the right, $\neg(P \vee Q)$: everything outside both circles.

Why should computer scientists care about Boolean algebra?

It is the hardware. Every CPU is Boolean algebra etched into silicon. Simplifying an expression *deletes physical gates*. The laws you can prove **are** the optimization.

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Machine learning: a decision tree is a Boolean formula. Each root-to-leaf path is a **conjunction**; a class is the **disjunction** of its paths:

$$\text{APPROVE} = (\text{age} < 30 \wedge \text{income} > 50\text{k}) \vee (\text{age} \geq 30 \wedge \text{score} > 700)$$

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And one Boolean function, **XOR**, is **not** computable by a single-layer perceptron — which is exactly why neural nets need hidden layers.

Group exercises

Random group assignments

Ashley, Isabelle: 7

Bailey, Tanna: 1

Blaisdell, Olyver: 10

Brown, Ashton: 9

Brown, Jonathan: 12

Cardenas, Dominic: 2

Cutler, George: 2

Fabik, Roland: 1

Fellows, Audrey: 7

Fitzgerald, Ryan: 11

Fuller, Ryan: 4

Griffen, Clara: 5

Grutsch, Tanner: 6

Hager, Nathan: 4

Horner, Jayden: 9

Johnson, Jaret: 8

Kintzel, Samantha: 8

Lameres, Alexis: 1

Lee, Holden: 8

Mayala, Reece: 11

McLean, Connor: 12

Mcbride, Daniel: 13

Miller, Maverick: 5

Moss, Emily: 6

Mousseau, Matthew: 3

Neuman, Evan: 2

Putnam, John: 3

Ramos, Vydel: 13

Rathbun, Jedidiah: 3

Roberts, Matthew: 6

Roller, Nathan: 5

Ryan, Otis: 10

Shevtsov, Timothy: 11

Sinclair, Aidan: 9

Smith, Charles: 4

Stensland, Emma: 12

Thiessen, Logan: 10

Thompson, Jaiden: 13

Vath, Ethan: 7

Group exercises

1. DeMorgan's laws are:

$$\neg(x \wedge y) = (\neg x) \vee (\neg y) \quad \text{and} \quad \neg(x \vee y) = (\neg x) \wedge (\neg y)$$

Prove the first of these (using truth tables). Then use DeMorgan's law to show how to disprove an if-and-only-if statement.

2. A **tautology** is a Boolean expression that evaluates to TRUE for all possible values of its variables. For example, the expression $x \vee \neg x$ evaluates to TRUE both when $x = \text{TRUE}$ and $x = \text{FALSE}$. Use truth tables to show the following are tautologies:

(a) $(x \vee y) \vee (x \vee \neg y)$

(b) $x \implies x$

(c) $\text{FALSE} \implies x$

(d) $(x \implies y) \wedge (y \implies z) \implies (x \implies z)$

3. A **contradiction** is a Boolean expression that evaluates to FALSE for all possible values of its variables. For example, the expression $x \wedge \neg x$ is a contradiction.

Use truth tables to show that the following are contradictions:

(a) $(x \vee y) \wedge (x \vee \neg y) \wedge \neg x$

(b) $x \wedge (x \implies y) \wedge (\neg y)$.

4. Reprove the items in #2 and #3 using the properties in Theorem 7.2 and the fact from Prop 7.3 that $x \implies y$ is equivalent to $(\neg x) \vee y$.

Solutions to group exercises

Solution to group exercise #1a

Problem. DeMorgan's laws are:

$$\neg(X \wedge Y) = (\neg X) \vee (\neg Y) \quad \text{and} \quad \neg(X \vee Y) = (\neg X) \wedge (\neg Y)$$

Prove the first of these (using truth tables).

Solution.

X	Y	$X \wedge Y$	$\neg(X \wedge Y)$	$\neg X$	$\neg Y$	$(\neg X) \vee (\neg Y)$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

The 4th and 7th columns have the same truth values. Hence,
 $\neg(X \wedge Y) = (\neg X) \vee (\neg Y)$.

Solution to group exercise #1b

Problem. Use DeMorgan's law to show how to disprove an if-and-only-if statement.

Solution. In Group Exercise #2 from the Theorems day, we showed that

$$A \iff B = (A \implies B) \wedge (B \implies A). \quad (1)$$

That is, A-if-and-only-if B is identical to if-A-then-B and if-B-then-A.

To **disprove** an if-and-only-if statement, we need to establish $\neg(A \iff B)$.
Now note that:

$$\begin{aligned} \neg(A \iff B) &= \neg\left((A \implies B) \wedge (B \implies A)\right) && \text{(by substituting Eq. (1)).} \\ &= \neg(A \implies B) \vee \neg(B \implies A) && \text{(by DeMorgan's law)} \end{aligned}$$

So we can disprove an if-and-only-if statement *either* by showing that $A \implies B$ fails *or* by showing that $B \implies A$ fails.

Solution to group exercise #2a

Problem. Use truth tables to show the following is a tautology:

$$(X \vee Y) \vee (X \vee \neg Y)$$

Solution.

X	Y	$\neg Y$	$X \vee Y$	$X \vee (\neg Y)$	$(X \vee Y) \vee (X \vee \neg Y)$
T	T	F	T	T	T
T	F	T	T	T	T
F	T	F	T	F	T
F	F	T	F	T	T

The last column shows that $(X \vee Y) \vee (X \vee \neg Y)$ is always true, and hence a tautology.

Solution to group exercise #2b

Problem. Use truth tables to show the following is a tautology:

$$X \implies X.$$

Solution. Recall the truth table for implication.

X	Y	$X \implies Y$
T	T	T
T	F	F
F	T	T
F	F	T

Here we have $Y=X$. So only the first and last row of the truth table ever occur. So $X \implies X$ is always true. Thus, $X \implies X$ is a tautology.

Solution to group exercise #2 c

Problem. Use truth tables to show the following is a tautology:

$$\text{FALSE} \implies X$$

Solution. The truth table for implication is

W	X	$W \implies X$
T	T	T
T	F	F
F	T	T
F	F	T

But here, we know W is FALSE. That is, only the last two rows of the truth table ever occur. So $\text{FALSE} \implies X$ is always true. Hence, $\text{FALSE} \implies X$ is a tautology.

Solution to group exercise #2 d

Problem. Use truth tables to show the following is a tautology:

$$(X \implies Y) \wedge (Y \implies Z) \implies (X \implies Z)$$

Solution.

Proposition	Nickname	Values							
X		T	T	F	F	T	T	F	F
Y		T	F	T	F	T	F	T	F
Z		T				F			
$X \implies Y$	A	T	F	T	T	T	F	T	T
$Y \implies Z$	B	T				F	T	F	T
$(X \implies Y) \wedge (Y \implies Z)$	$C := A \wedge B$	T	F	T	T	F	F	F	T
$X \implies Z$	D	T				F	F	T	T
$(X \implies Y) \wedge (Y \implies Z) \implies (X \implies Z)$	$C \implies D$	T				T			

Solution to group exercise #3a

Problem. Use truth tables to show the following is a contradiction:

$$(X \vee Y) \wedge (X \vee \neg Y) \wedge \neg X$$

Solution.

Proposition	Nickname	Truth Values			
X		T	T	F	F
Y		T	F	T	F
$X \vee Y$	A	T	T	T	F
$\neg Y$		F	T	F	T
$X \vee (\neg Y)$	B	T	T	F	T
$\neg X$	C	F	F	T	T
$(X \vee Y) \wedge (X \vee \neg Y) \wedge \neg X$	$A \wedge B \wedge C$	F	F	F	F

Solution to group exercise #3b

Problem. Use truth tables to show the following is a contradiction:

$$X \wedge (X \implies Y) \wedge (\neg Y)$$

Solution.

Proposition	Nickname	Truth Values			
X	A	T	T	F	F
Y		T	F	T	F
$X \implies Y$	B	T	F	T	T
$\neg Y$	C	F	T	F	T
$X \wedge (X \implies Y) \wedge (\neg Y)$	$A \wedge B \wedge C$	F	F	F	F

Solution to group exercise #4

The solutions to group exercise #4 refer to the properties from the textbook below.

Theorem 7.2

- $x \wedge y = y \wedge x$ and $x \vee y = y \vee x$. (Commutative properties)
- $(x \wedge y) \wedge z = x \wedge (y \wedge z)$ and $(x \vee y) \vee z = x \vee (y \vee z)$. (Associative properties)
- $x \wedge \text{TRUE} = x$ and $x \vee \text{FALSE} = x$. (Identity elements)
- $\neg(\neg x) = x$.
- $x \wedge x = x$ and $x \vee x = x$.
- $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$ and $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$. (Distributive properties)
- $x \wedge (\neg x) = \text{FALSE}$ and $x \vee (\neg x) = \text{TRUE}$.
- $\neg(x \wedge y) = (\neg x) \vee (\neg y)$ and $\neg(x \vee y) = (\neg x) \wedge (\neg y)$. (DeMorgan's Laws)

Figure 3: Boolean Algebra Properties

Solution to group exercise #4 - Redo of 2a

Problem. Show the following is a tautology:

$$(X \vee Y) \vee (X \vee \neg Y)$$

Solution.

$$\begin{aligned}(X \vee Y) \vee (X \vee \neg Y) &= (X \vee X) \vee (Y \vee \neg Y) && \text{(commutative, associative props.)} \\ &= X \vee \text{True} && \text{(unnamed props \#5,7)} \\ &= \text{True} && \text{(unnamed prop \#7)}\end{aligned}$$

Solution to group exercise #4 - Redo of #2b

Problem. Show the following is a tautology:

$$X \implies X.$$

Solution.

$$\begin{aligned} X \implies X &= (\neg X \vee X) && \text{(Prop 7.3)} \\ &= \text{True} && \text{(unnamed prop \#7)} \end{aligned}$$

Solution to group exercise #4 - Redo of #2 c

Problem. Show the following is a tautology:

$$\text{FALSE} \implies X$$

Solution.

$$\begin{aligned}\text{FALSE} \implies X &= (\neg \text{FALSE}) \vee X && \text{(Prop 7.3)} \\ &= \text{TRUE} \vee X \\ &= \text{TRUE}\end{aligned}$$

Solution to group exercise #4 - Redo of #2 d

Problem. Show the following is a tautology:

$$(X \implies Y) \wedge (Y \implies Z) \implies (X \implies Z)$$

$$\underbrace{(X \implies Y) \wedge (Y \implies Z)}_{:= A} \implies \underbrace{(X \implies Z)}_{:= B}$$

$$= \neg A \vee B \quad (\text{Prop 7.3})$$

$$= \left(\neg[X \implies Y] \vee [\neg(Y \implies Z)] \right) \vee (X \implies Z) \quad (\text{DeMorgan's Law})$$

$$= (X \vee \neg Y) \vee (Y \vee \neg Z) \vee (\neg X \vee Z) \quad (\text{Prop 7.3, DeMorgan's Law})$$

$$= (X \vee \neg X) \vee (Y \vee \neg Y) \vee (Z \vee \neg Z) \quad (\text{Associative, commutative props.})$$

$$= \text{TRUE} \vee \text{TRUE} \vee \text{TRUE} \quad \text{Unnamed Prop \#7}$$

$$= \text{TRUE}$$

Solution to group exercise #4 - Redo of #3a

Problem. Show the following is a contradiction:

$$(X \vee Y) \wedge (X \vee \neg Y) \wedge \neg X$$

Solution.

$$\begin{aligned} & (X \vee Y) \wedge (X \vee \neg Y) \wedge \neg X \\ \Rightarrow & (X \wedge X \wedge \neg X) \vee (X \wedge \neg Y \wedge \neg X) \vee (Y \wedge X \wedge \neg X) \vee (Y \wedge \neg Y \wedge \neg X) && \text{Distributive prop.} \\ \Rightarrow & \text{FALSE} \vee \text{FALSE} \vee \text{FALSE} \vee \text{FALSE} && \text{Unnamed prop. \# 7} \\ \Rightarrow & \text{FALSE} \end{aligned}$$

Remark

A tricky part of applying the properties is using the distributive law correctly. For intuition, recall how multiplication distributes over addition [e.g. $4(3 + 5) = 4 \cdot 3 + 4 \cdot 5$].

Solution to group exercise #4 - Redo of #3b

Problem. Show the following is a contradiction:

$$X \wedge (X \implies Y) \wedge (\neg Y)$$

Solution.

$$\begin{aligned} & X \wedge (X \implies Y) \wedge (\neg Y) \\ =& X \wedge (\neg X \vee Y) \wedge (\neg Y) \\ =& (X \wedge \neg X \wedge \neg Y) \vee (X \wedge Y \wedge \neg Y) \\ =& \text{FALSE} \vee \text{FALSE} \\ =& \text{FALSE} \end{aligned}$$

Prop 7.3

Distributive prop.

Unnamed prop. # 7